

In-Class Exercise Sheet 5

In the upcoming lab session, we will discuss reproducible builds. In this lecture, we recap some bare basics, to make sure everybody is on the same page: Reading simple C-programs, building C-programs, role of the preprocessor, and Make files.

1.



Code Magnets

The College Blackjack Team was working on some code on the dorm fridge, but someone mixed up the magnets! Can you reassemble the code from the magnets?

Enter two characters
for the card name.

```

/*
 * Program to evaluate face values.
 * Released under the Vegas Public License.
 * (c)2014 The College Blackjack Team.
 */

.....

.....

.....main()
{
    char card_name[3];
    puts("Enter the card_name: ");
    scanf("%2s", card_name);
    int val = 0;
    if (card_name[0] == 'K') {
        val = 10;
    } else if (card_name[0] == 'Q') {

        .....

    } else if (card_name[0] == ..... ) {
        val = 10;

    } ..... (card_name[0] == ..... ) {

        .....

    } else {
        val = atoi(card_name);
    }
    printf("The card value is: %i\n", val);
    ..... 0;
}

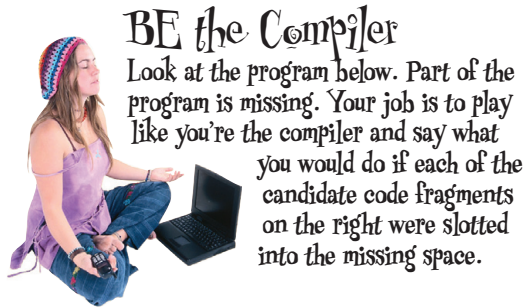
```

<stdlib.h> ;
 ; val = 11
 int 'J'
 #include 'A'

return
 else #include
 if val = 10
 <stdio.h>

From the book “Head First C” by David Griffiths. Code available from <https://github.com/dogriffiths/HeadFirstC>.

2.



Candidate code goes here.

```
#include <stdio.h>

printf("A day on Mercury is %f hours\n", day);
return 0;
}

float mercury_day_in_earth_days()
{
    return 58.65;
}

int hours_in_an_earth_day()
{
    return 24;
}
```

Here are the code fragments.

```
float mercury_day_in_earth_days();
int hours_in_an_earth_day();

int main()
{
    float length_of_day = mercury_day_in_earth_days();
    int hours = hours_in_an_earth_day();
    float day = length_of_day * hours;
```

Mark the boxes that
you think are correct.

- ☐ You can compile the code.
- ☐ You should display a warning.
- ☐ The program will work.

```
float mercury_day_in_earth_days();

int main()
{
    float length_of_day = mercury_day_in_earth_days();
    int hours = hours_in_an_earth_day();
    float day = length_of_day * hours;
```

- ☐ You can compile the code.
- ☐ You should display a warning.
- ☐ The program will work.

```
int main()
{
    float length_of_day = mercury_day_in_earth_days();
    int hours = hours_in_an_earth_day();
    float day = length_of_day * hours;
```

- ☐ You can compile the code.
- ☐ You should display a warning.
- ☐ The program will work.

```
float mercury_day_in_earth_days();
int hours_in_an_earth_day();

int main()
{
    int length_of_day = mercury_day_in_earth_days();
    int hours = hours_in_an_earth_day();
    float day = length_of_day * hours;
```

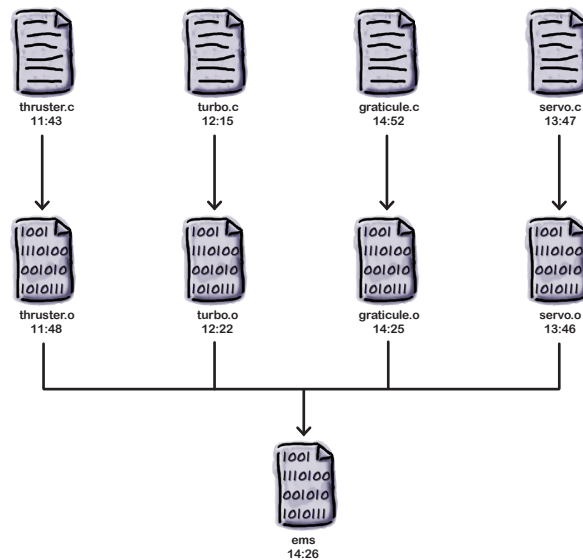
- ☐ You can compile the code.
- ☐ You should display a warning.
- ☐ The program will work.

From the book “Head First C” by David Griffiths.

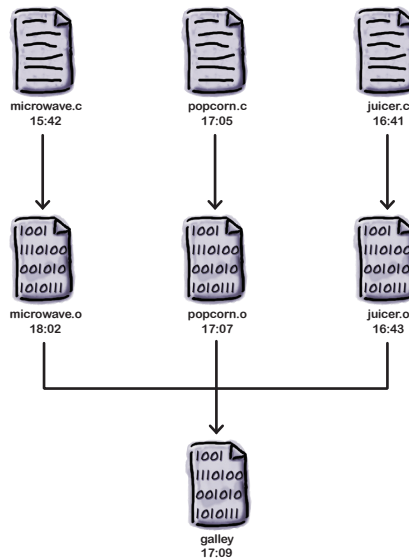
3.

**Long Exercise**

Here is some of the code that's used to control the engine management system on the craft. There's a timestamp on each file. Which files do you think need to be recreated to make the `ems` executable up to date? Circle the files you think need to be updated.



And in the galley, they need to check that their code's up to date as well. Look at the times against the files. Which of these files need to be updated?



From the book “Head First C” by David Griffiths. Code available from <https://github.com/dogriffiths/HeadFirstC>.



Make Magnets

Hey, baby, if you don't groove to the latest tunes, then you'll *love* the program the guys in the Head First Lounge just wrote! `oggswing` is a program that reads an Ogg Vorbis music file and creates a swing version. Sweet! See if you can complete the makefile that compiles `oggswing` and then uses it to convert a `.ogg` file:

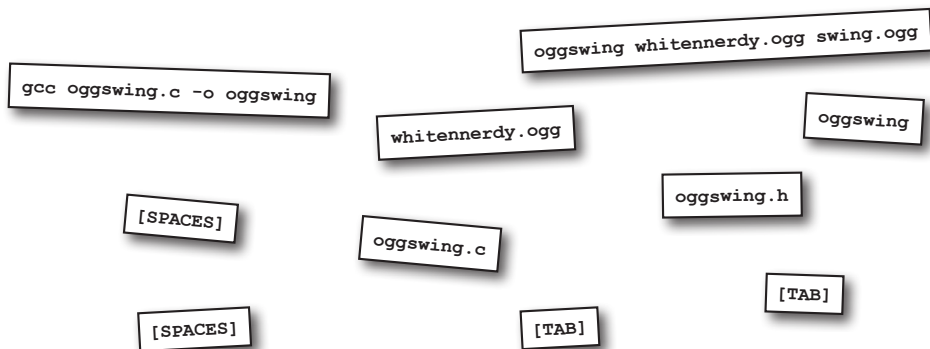
This converts
`whitennerdy.ogg`
to `swing.ogg`.

`oggswing:`

.....

`swing.ogg:`

.....



From the book “Head First C” by David Griffiths. Code available from <https://github.com/dogriffiths/HeadFirstC>.

5. The following Makefile causes unreproducible builds:

```
1 # Collect all C source files in the current directory.
2 #
3 # $(wildcard *.c) finds all matching C source files.
4 SRCS = $(wildcard *.c)
5
6 # Build the executable "tool" from the corresponding object files.
7 tool: $(SRCS:.c=.o)
8     $(CC) -o $@ $^
```

What causes the problem?

How could you fix the problem?

Adapted from Reproducible Builds website (<https://reproducible-builds.org>).

6. How does the C preprocessor process the following macros? State the output.

(a)

```
1 #define BUFFSIZE 1024
2 int buf[BUFFSIZE + 1];
```

(b)

```
1 #define a(b) b + 1
2 int x = a(1) + 1;
```

(c) Contents of file testCPP.c:

```
1 #include <stdio.h>
2 int main(void) {
3     printf("__LINE__ = %d\n", __LINE__);
4     printf("__FILE__ = %s\n", __FILE__);
5     printf("__TIME__ = %s\n", __TIME__);
6     printf("__DATE__ = %s\n", __DATE__);
7
8     return 0;
9 }
```

Last example from the book “Programming in C: A Practical Approach” by Ajay Mittal.

7. Consider the following C-program:

```
1  #include <assert.h> // Include and enable assertions.
2  #include <stdio.h>
3
4  #define FALSE 0
5
6  char *p = (char *)5;
7
8  int someinitialization(void)
9  {
10     p = "abc";
11     return FALSE;
12 }
13
14 int main(int argc, char **argv)
15 {
16     assert(someinitialization() == FALSE);
17
18     printf("%s\n", p);
19
20     return 0;
21 }
```

Calling “gcc heisenbug.c -o heisenbug; ./heisenbug” yields _____,

“gcc -DDEBUG heisenbug.c -o heisenbug; ./heisenbug” yields _____.

What causes this heisenbug?

Example from <https://jacquesmattheij.com/a-c-heisenbug-in-the-wild/>.